

CS 2050 Technical Documentation

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General Resources

Your first source should always be:

- Your class lecture slides, documents and the resources linked in the canvas materials, instructor (Deb), learning assistant (LA)
-

Tools:

- [Visualize Java Code](#) - used to step through code and edit code
-

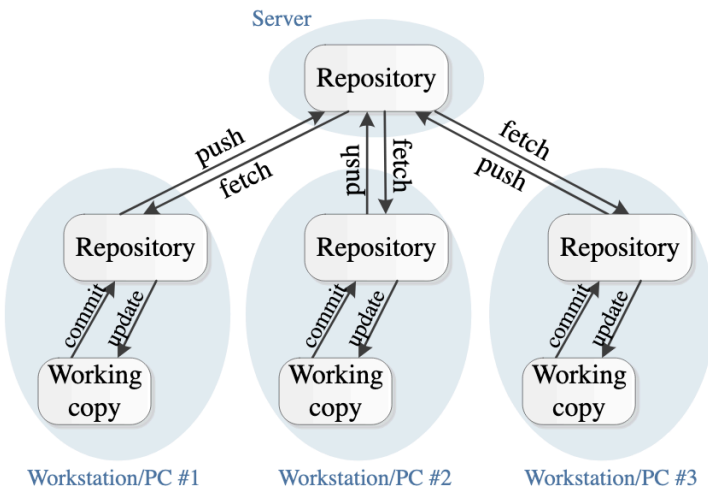
Feel free to add more resources shared from the lectures.

Development Environment

Version Control

A system that tracks and manages changes to files over time. A software tool that organizes every version I create, like code. It keeps a history of changes where it allows me to revert to previous versions. Each version is an update of the file or code.

Distributed version control in git



<https://www.google.com/url?sa=i&url=https%3A%2F%2Fhomes.cs.washington.edu%2F~mernst%2Fadvice%2Fversion-control.html>

Integrated Development Environment (IDE)

A Software application that provides programmers all the tools for software development: debugger, write code, build automation tools (compile and run projects), and most have syntax highlighting. For example: Eclipse, Visual Studio Code (VS), PyCharm, NetBeans, Xcode, etc.

Git and GitHub

Git and Github are different but related tools used in software development for version control and collaboration.

Git

- Version control system
- Track changes of codes and other files over time
- Let's you go back to projects if needed
- Works offline

GitHub

- Stores Git repositories online, making them accessible with others
- Provides collaborations tools (pull requests for code view, issues, project boards)
- Needs internet to use
- GitHub is a service that hosts Git repositories

- Local git repo with Screenshot of your local git repository folders and files
- github desktop version control steps with commands and their description
- Github remote repository with screenshot of Eclipse Project Folder and java files
- Gitignore file
- Resource List to help you

Eclipse IDE

Summarize Eclipse and include resources that will help you use eclipse.

Module 1

Note* **Algorithm**: set of steps to complete a task. An algorithm can be used to solve a problem. An Algorithm has 3 main parts.

- Input: Having data or information to give to the algorithm.
- Process: A series of logical and arithmetic steps that the algorithm performs on the input.
- Output: The results or conclusion that the algorithm produces after completing its process.

Classes

Classes are constructs that define objects with the same data type. A class also provides “special” types of methods, known as constructors, which are invoked (initiate execution of a function like a method.) to construct objects from the class.

- Class is not a real world entity. It is only a blueprint or prototype from which objects are created.
- Class can have groups of variables of different types...
- Class can have variety of methods to define behaviors
- A class is a plan that defines common attributes and behaviors that all the objects created from it will share.

Objects

An object that is created as an instance from a class. An object has two important pieces.

- **Behavior**
 - Things an object **does**
 - Methods that operate on the object

- For instance, a “car” object may have behaviors like lock doors, honk, turn on/off engine and lights, brake, accelerate, etc.
- **State**
 - Things an object **knows**
 - The data fields (attributes) that define an object
 - For example, a “car” object may have attributes like color, model, year, gear, fuel level, etc.

Constructors

Constructors can be used to initialize values to the data member fields of a class when creating an object.

- A constructor has the exact **name** as the class
- A constructor can overload, meaning it can hold multiple constructors with different parameters
- There is **no** return type. Including void.
- It is **not** necessary to write a constructor in a class

Instance Variables & Instance Methods

Instance Variable

- A variable that is part of an instance within a class
- State of an object: things an object **knows** about itself

```
// Athlete class
class Athlete {

    // Instance variables
    private String firstName;
    private String lastName;
    private double weight;
    private double height;
    private int age;
    private int[] dailyCaloriesBurned;
```

This screenshot shows instance variables for being created in a class.

*Why make it private? So no other programmer can change the objects

Instance Method

- A method that is part of an instance of a class
- These can only be executed through a reference to an instance
- Behavior of an object: things an object **can do**

2D Arrays

A two-dimensional array is like a table, matrix or spreadsheet that are arranged in **rows and columns**.

- A 2D array has an array inside of an array!

```
//Constant variables to be absolute
final int TABLE_ROW_SIZE = 5;
final int TABLE_COL_SIZE = 6;

//Declaring a 2D Array named table
int[][] table = new int [TABLE_ROW_SIZE] [TABLE_COL_SIZE];

/*Nested for loop
 *Outer loop goes through each row
 *Inner loop goes through each column for CURRENT row
 */
for (int row = 0; row < TABLE_ROW_SIZE; row++)
{
    for (int col = 0; col < TABLE_COL_SIZE; col++)
    {
        table[row][col] = (row + 1) * (col + 1);
    }
} // End loop
```

Heading 2

Add heading for different sections